Inspecting failures and anomalies

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

Investigate unusual outcomes without erasing valid records

Separate valid slow behavior, demonstrated measurement errors and untested explanations.

Keep one habit: Question and contract → complete main ledger → separate picture → boundary probes → investigation → independent attempt.

MAIN LESSON · SECTION 1

Investigate unusual outcomes without erasing valid records

> Separate valid slow behavior, demonstrated measurement errors and untested explanations.

Learning goals

I can explain the **mechanism** and terminology of inspecting failures and anomalies.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Retrieve before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Running the primary experiment. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Power and resource tradeoffs. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Privacy and data governance. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Starting explanation

- Recall Running the primary experiment. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**

End goal

Explain inspecting failures and anomalies using the records and boundaries of this fictional investigation.

▼ Earlier wording, alternatives and picture transcripts

Your learning target

Remember before reading

This week you will practise:

Explain the **mechanism** and terminology of inspecting failures and anomalies

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

This week’s end goal

MAIN LESSON · SECTION 2

Understand the question and evidence contract

> Separate valid slow behavior, demonstrated measurement errors and untested explanations.

Start with the research problem

An unusual value can reveal a real failure mode. Before changing it, inspect the input, raw trace and planned inclusion rule. A demonstrable unit or capture error can justify a documented correction; an inconvenient valid result cannot be deleted just because it changes the graph.

Keep post-result explanation searches separate from the original primary analysis, and distinguish a logged event from a complete causal diagnosis.

Original concept explanation

Failure inspection generates explanations to test. An unusual result is not automatically a bad data point.

Complete case inputs, rules and limits

Anomaly item	Complete supplied contract
Main valid durations	1, 2, 2, 15 seconds; all stipulated valid traces
Proposed main deletion	Remove fifteen seconds without a rule
Main analysis	Retain all four in the declared primary summary
Separate picture durations	2, 2, 2, 10 seconds
Picture log	Fourth run records a source timeout; plan includes timeouts
Correction requirement	Evidence of a measurement error and preserved provenance, not dislike of an outcome
Scope	Supplied timing traces; no actual timing, timeout generation or root-cause experiment

Name given and missing evidence

List the given facts and one fact that is not supplied.

▼ **Earlier wording, alternatives and picture transcripts**

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

MAIN LESSON · SECTION 3

Trace the complete main case before interpreting it

> Follow each evidence-to-conclusion step and retain unknown quantities.

Original declared procedure

Compare raw **output**s, **input** validity and procedure traces. Correct demonstrable measurement errors with provenance; retain valid unusual **observation**s. Label post-result investigations exploratory.

Complete main-case evidence ledger

Main calculation	Complete arithmetic	Interpretation
Retained durations	1, 2, 2, 15 seconds	All valid under supplied contract
Sum / count	20 seconds / 4	No undeclared exclusion
Mean	20/4 = 5 seconds	Sensitive to the long valid duration
Median	Middle values 2 and 2; average 2 seconds	Different descriptive aspect
Wrong-deletion illustration	(1+2+2)/3 = 5/3 seconds	Shows distortion; not a replacement primary result
Cause of fifteen-second duration	Not supplied	Propose an investigation without inventing its outcome

State the scope and explain the visual

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Earlier wording, alternatives and picture transcripts

Inspecting failures and anomalies — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Durations 1, 2, 2, 15 seconds are all valid traces. Team deletes 15 as an outlier without a rule.	Compare raw outputs, input validity and procedure traces. Correct demonstrable measurement errors with provenance; retain valid unusual observations.	The slow valid case must remain in the declared primary analysis. Investigate mechanism separately; mean 5 and median 2 reveal different aspects.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Connect the picture and the explanation

MAIN LESSON · SECTION 4

Predict the changed case before reading the explanation

> Keep the rule and evidence boundary visible.

Main task and response

- Durations 1, 2, 2, 15 seconds are all valid traces. Team deletes 15 as an outlier without a rule. Show the intermediate steps and give the conclusion with its scope.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Predict the pictured change

- Change: Delete the 10-second record only because it looks bad.

▼ **Check after predicting**

Prediction: The mean becomes 2 seconds but the selective report violates this plan.

Reason: Observed inconvenient outcomes are not an approved exclusion reason.

▼ **Earlier wording, alternatives and picture transcripts**

Compare a familiar case with a changed case

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Connect the main result with the separate pictured case

> Keep the two scenarios and their supplied evidence distinct.

Main case and original checked reasoning

Durations 1, 2, 2, 15 seconds are all valid traces. Team deletes 15 as an outlier without a rule.

The slow valid case must remain in the declared primary analysis. Investigate **mechanism** separately; mean 5 and median 2 reveal different aspects.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Separate pictured inputs and complete explanation

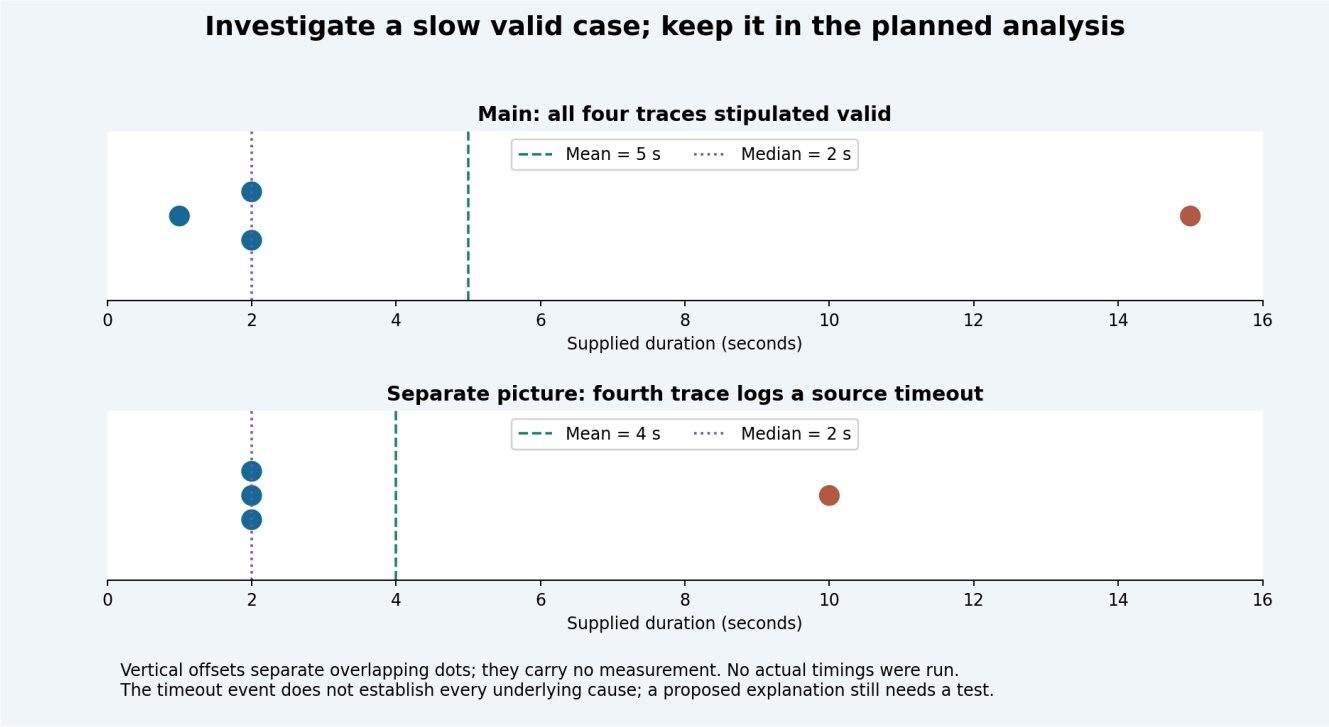
Inspect an **anomaly** without erasing it. A timing study records four runs and notices one unusually long duration.

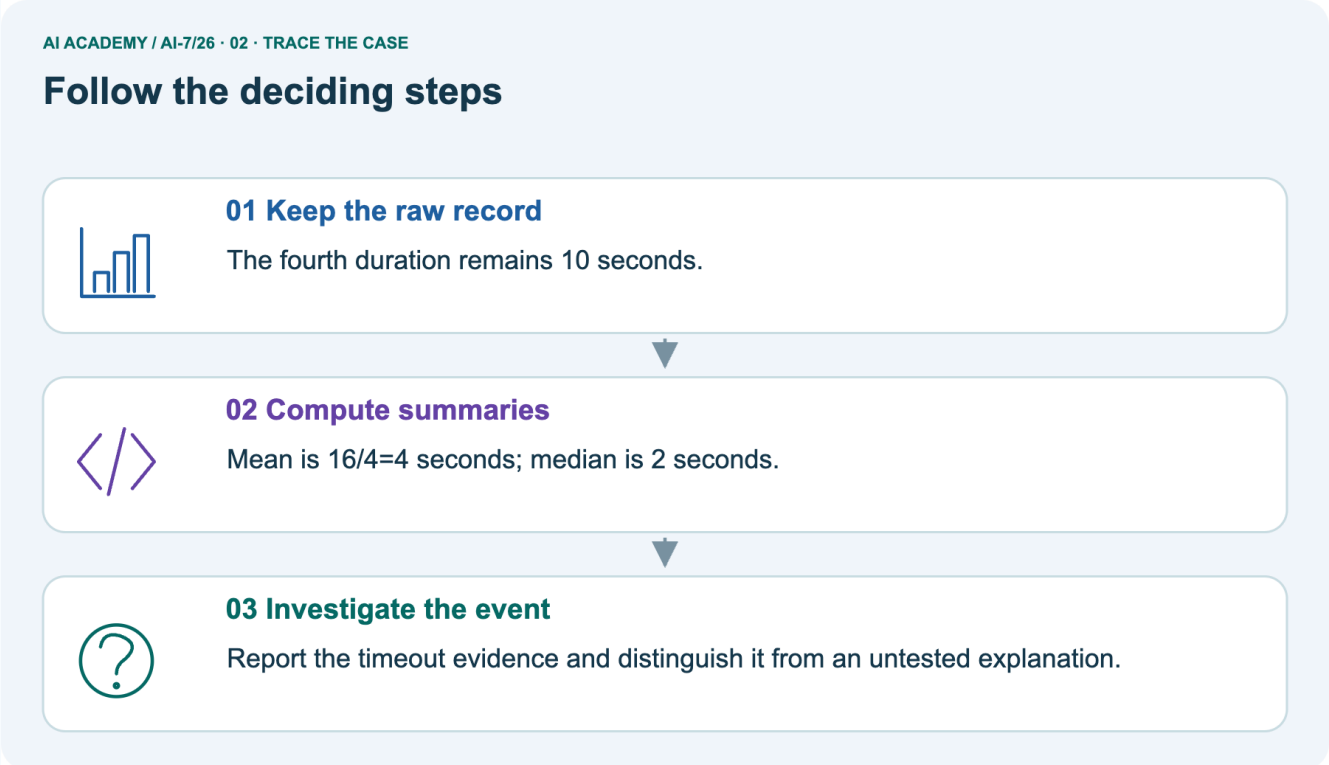
Durations are 2,2,2,10 seconds. Logs show a source timeout on the fourth run. The plan includes timeouts in reporting.

The separate picture records 2, 2, 2 and 10 seconds. The sum is sixteen, mean four and median two seconds. The fourth log records a source timeout, and the predefined reporting plan includes timeouts. That record remains in the primary summary. Deleting it merely because it looks bad would produce a two-second mean and conceal the stipulated slow case. A timeout event does not identify every underlying cause.

Inspect trace boundaries, source behavior and known conditions, then design a controlled follow-up; label proposed explanations and follow-ups as untested until evidence exists.

Read the labelled representation





Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Keep the raw record	The fourth duration remains 10 seconds.
2. Compute summaries	Mean is 16/4=4 seconds; median is 2 seconds.
3. Investigate the event	Report the timeout evidence and distinguish it from an untested explanation.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

An **anomaly** invites investigation, not automatic deletion. A slow failure may be part of real service behaviour; the predefined rule determines inclusion.

Complete picture evidence and limits

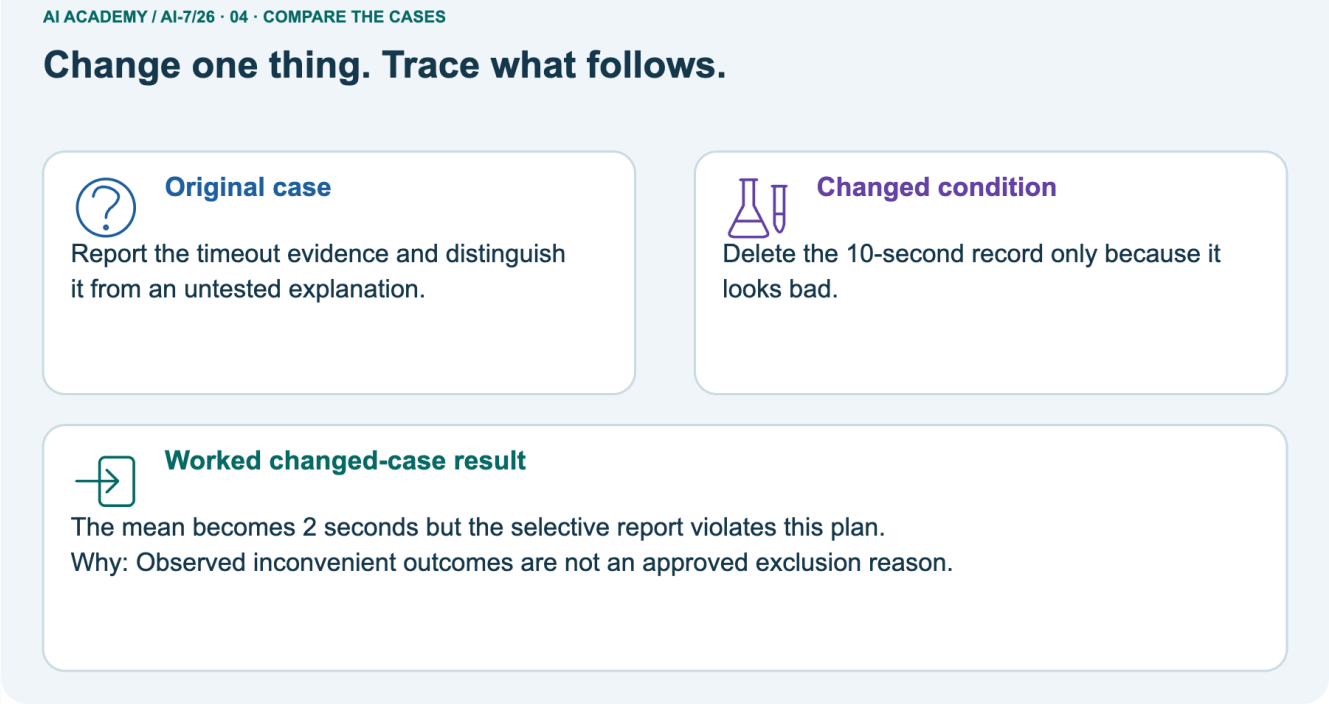
Separate picture quantity	Supported record
Durations	2, 2, 2, 10 seconds
Mean	16/4 = 4 seconds
Median	(2+2)/2 = 2 seconds
Fourth log	Source timeout recorded
Plan	Includes timeouts in reporting

Separate picture quantity	Supported record
Delete ten because it looks bad	Violates plan; hypothetical mean falls to two
Root cause	Not fully identified by timeout event
Follow-up status	Proposed exploratory investigation, not a completed test

Predict the changed condition

- Change one thing: Delete the 10-second record only because it looks bad. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain what remains unknown

- A logged timeout does not identify every underlying cause; report conditions and test explanations.

- Read the labelled picture: Calculate mean and median for 2,2,2,10. What does the timeout log establish, and why is deleting that inconvenient record incompatible with the stated plan?

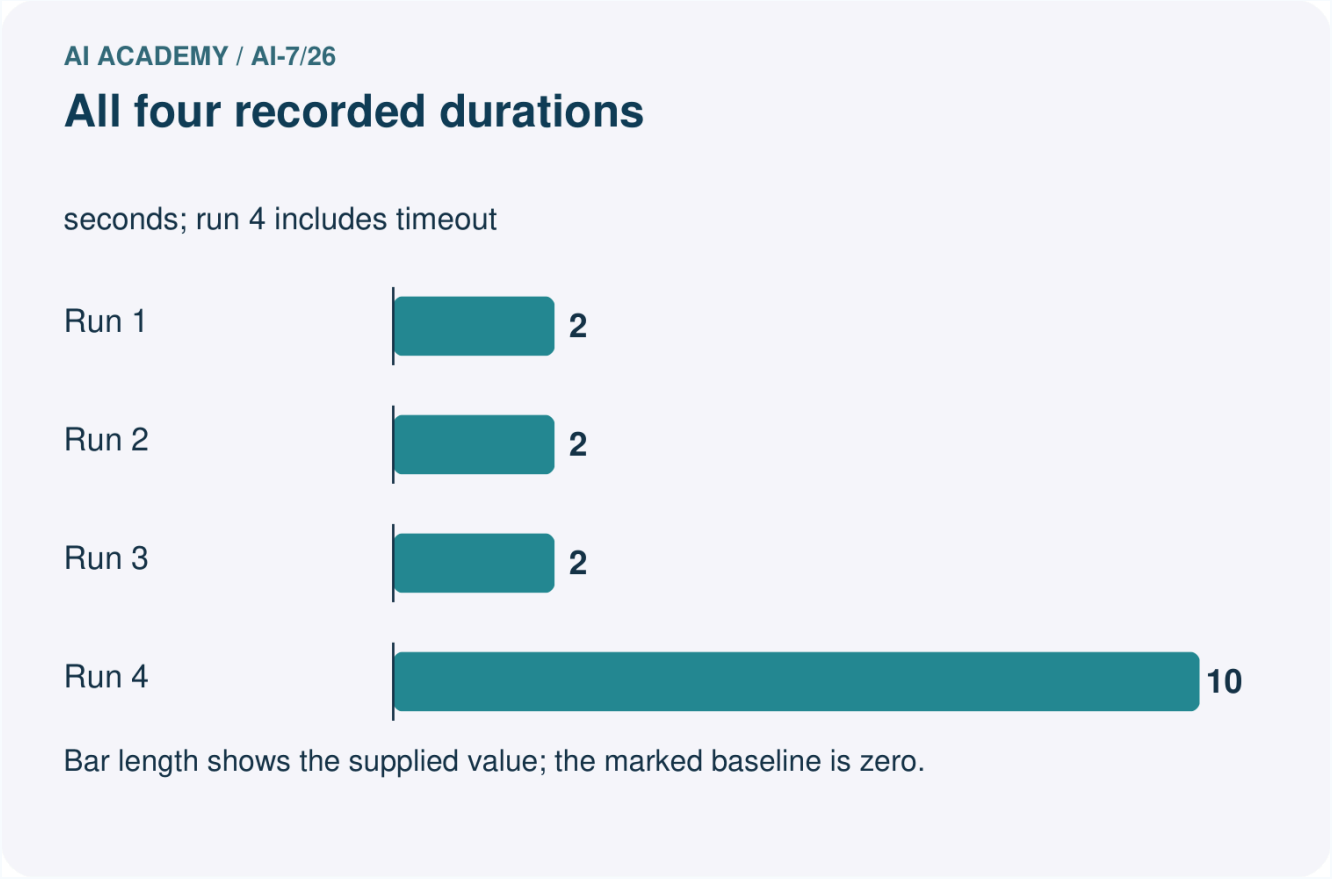
▼ Earlier wording, alternatives and picture transcripts

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Inspecting failures and anomalies

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/26

Inspecting failures and anomalies

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Keep the raw record

The fourth duration remains 10 seconds.





2. Compute summaries

Mean is 16/4=4 seconds; median is 2 seconds.





3. Investigate the event

Report the timeout evidence and distinguish it from an untested explanation.

CHANGE ONE THING

Delete the 10-second record only because it looks bad.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Find the first unsupported inference and repair it

> Use the supplied conditions rather than inventing missing evidence.

Original qualification and repair setup

Post-hoc exclusion based on an unwanted outcome can distort the conclusion.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Complete boundary and misconception probes

Anomaly audit	Checked interpretation
Value unusually large	Not automatically invalid data
All traces stipulated valid	Keep them under the declared inclusion rule
Mean differs from median	Different summaries; neither licenses deleting the tail
Log shows timeout	Event evidence, not a complete causal explanation
Known unit conversion error demonstrated	Correct transparently with original record and evidence
No evidence of capture error	Do not invent one to improve results
Follow-up inspired by outcomes	Label exploratory and preserve original analysis
Wrong-deletion calculation shown	Illustration of distortion, not a new approved result

Write and justify the repair

- A report omits this qualification: Post-hoc exclusion based on an unwanted outcome can distort the conclusion. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

State one repair and the checks you would rerun.

▼ Earlier wording, alternatives and picture transcripts

Locate the earliest unsupported step

MAIN LESSON · SECTION 8

Practise the mechanism with progressively less help

Keep the selected independent case for a first attempt.

Shared explanation

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

> **Explain your prediction to a partner before comparing results.**

Supported workbook trace

Use workbook W2 for the connected example “Inspecting failures and anomalies”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual scope check

- Use the worked case for Inspecting failures and anomalies. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Picture response

- Calculate mean and median for 2,2,2,10. What does the timeout log establish, and why is deleting that inconvenient record incompatible with the stated plan?

> **My explanation using a pictured step and the given facts**

Reserve the independent task

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

Practise with less help: Trace the worked case

Check your reasoning

Explain the picture

Investigate, record the evidence and check the limits

> Distinguish a prediction, a paper trace and an actual local execution.

Current investigation

Durations 1, 2, 2, 15 seconds are all valid traces. Team deletes 15 as an outlier without a rule. Show the intermediate steps and give the conclusion with its scope.

Complete investigation instructions

1. Predict mean five and median two for the main valid traces. The complete exact arithmetic retains those records and separately summarizes the picture as mean four, median two. Its explicitly hypothetical deletion summaries show how the result would be distorted; they never overwrite the primary record. Record what the source-timeout log establishes and what it leaves unknown.

   Propose a controlled follow-up with fixed inputs and clear timing boundaries, but keep its status not run. The program performs no actual timings or network calls. Reserve the independent 2, 3, 4, 20 case for the first attempt.

```
"""Describe supplied valid durations without erasing inconvenient observations."""
from fractions import Fraction

def timing_summary(values):
 if not isinstance(values,(list,tuple)) or not values or any(type(v) is not int or v<0 for
v in values):raise ValueError('nonempty nonnegative integer durations required')
 ordered=sorted(values);n=len(values)
 middle=Fraction(ordered[n//2]) if n%2 else Fraction(ordered[n//2-1]+ordered[n//2],2)
 return {'count':n,'mean_seconds':Fraction(sum(values),n),'median_seconds':middle}
main_values=[1,2,2,15];before=main_values[:];main=timing_summary(main_values)
picture_values=[2,2,2,10];picture=timing_summary(picture_values)
# Wrong-deletion illustration only, never replaces either retained record.
hypothetical_main_deleted=timing_summary(main_values[:-1])
hypothetical_picture_deleted=timing_summary(picture_values[:-1])
assert main['mean_seconds']==5 and main['median_seconds']==2 and main_values==before
assert picture['mean_seconds']==4 and picture['median_seconds']==2
print('Retained main valid durations:',main_values,main)
print('Retained separate picture durations:',picture_values,picture,'fourth trace logs a
source timeout')
print('Illustrative wrong deletion would change main mean
```

```
to',hypothetical_main_deleted['mean_seconds'],'and picture mean
to',hypothetical_picture_deleted['mean_seconds'])
print('These deletion summaries demonstrate distortion only; original valid observations
and the planned timeout inclusion are retained.')
print('A timeout log is a recorded event, not a complete root-cause diagnosis. A genuine
measurement correction needs evidence and provenance, not dislike of an outcome.')
print('No response is timed, no timeout is generated, and no causal explanation or
population tail estimate is established.')
```

Evidence record and reflection

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

Predict, investigate and preserve the evidence

Plan → try → check

Try the selected new case independently

> Calculate both summaries, retain valid observations and distinguish a proposed explanation from evidence.

Complete independent task

Values 2, 3, 4, 20 are valid. Compute mean and median and propose an honest investigation.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Apply a new case independently

Attempt this without hints

Values 2, 3, 4, 20 are valid. Compute mean and median and propose an honest investigation.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

MAIN LESSON · SECTION 11

Save the research artifact and unresolved conditions

> Keep supplied evidence, proposed revisions and actual work distinct.

Project task and artifact

Add a labelled inspecting failures and anomalies **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of inspecting failures and anomalies.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence and limitations

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

▼ **Earlier wording, alternatives and picture transcripts**

Evidence for your project

MAIN LESSON · SECTION 16

Retrieve the mechanism and state the evidence boundary

> Explain what follows and what would require new evidence.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of inspecting failures and anomalies

> **My deciding step, evidence and one remaining question**

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use research vocabulary precisely**

REFERENCE · SECTION 5

Reference · Use research vocabulary precisely

> Connect each term to the question, comparison, measurement or evidence.

Vocabulary and your explanation

Word	Meaning
Anomaly	An unusual observation requiring investigation
Provenance-backed correction	A repair justified by source evidence
Exploratory analysis	A follow-up chosen after results

Use two terms to explain a step in this week's example.

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

- A report omits this qualification: Post-hoc exclusion based on an unwanted outcome can distort the conclusion. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ Reference · Check the research reasoning and its limits

REFERENCE · SECTION 12

Reference · Check the research reasoning and its limits

> Use after practice; reserve the selected independent case for its first attempt.

Current explanation and response

- Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

  For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

My limited claim and a useful next check

▼ **Extra practice · Revisit the complete trace**

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

Inspecting failures and anomalies — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Durations 1, 2, 2, 15 seconds are all valid traces. Team deletes 15 as an outlier without a rule.	Compare raw outputs, input validity and procedure traces. Correct demonstrable measurement errors with provenance; retain valid unusual observations.	The slow valid case must remain in the declared primary analysis. Investigate mechanism separately; mean 5 and median 2 reveal different aspects.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: Inspect an **anomaly** without erasing it. A timing study records four runs and notices one unusually long duration.

By the end, I can	How I will show it
Explain the mechanism and terminology of inspecting failures and anomalies	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Inspect an **anomaly** without erasing it. A timing study records four runs and notices one unusually long duration.
- Which labelled step matches this detail? Report the timeout **evidence** and distinguish it from an untested explanation.
- 1. Investigate the event
- 2. Keep the raw record
- 3. Compute summaries
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Investigate the event Report the timeout **evidence** and distinguish it from an untested explanation. An **anomaly** invites investigation, not automatic deletion.

A slow failure may be part of real service behaviour; the predefined rule determines inclusion.

Changed-case question and optional hint

- Changed-case practice: Delete the 10-second record only because it looks bad. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. The mean becomes 2 seconds but the selective report violates this plan.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: The mean becomes 2 seconds but the selective report violates this plan. Observed inconvenient outcomes are not an approved exclusion reason.

Repair and retain the record

Review Running the primary experiment (ai-7/25). Goal: Explain the **mechanism** and terminology of running the primary experiment. Short practice: Compute A's and B's six-case counts and identify the cases that favour each.

How must a B revision tuned on case four be labelled when evaluated again on the same set?. Return to this week and explain what you changed.

Review Power and resource tradeoffs (ai-7/23). Goal: Explain the **mechanism** and terminology of power and resource tradeoffs. Short practice: Calculate both simulated detection fractions and resource costs.

What additional detection is observed for the extra forty units, and does that establish power for an actual classroom study?. Return to this week and explain what you changed.

Review Privacy and data governance (ai-7/20). Goal: Explain the **mechanism** and terminology of privacy and data governance. Short practice: Point to the analysis table and the restricted identity key.

Why are the codes pseudonymous rather than guaranteed anonymous, and what changes if the key is published?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

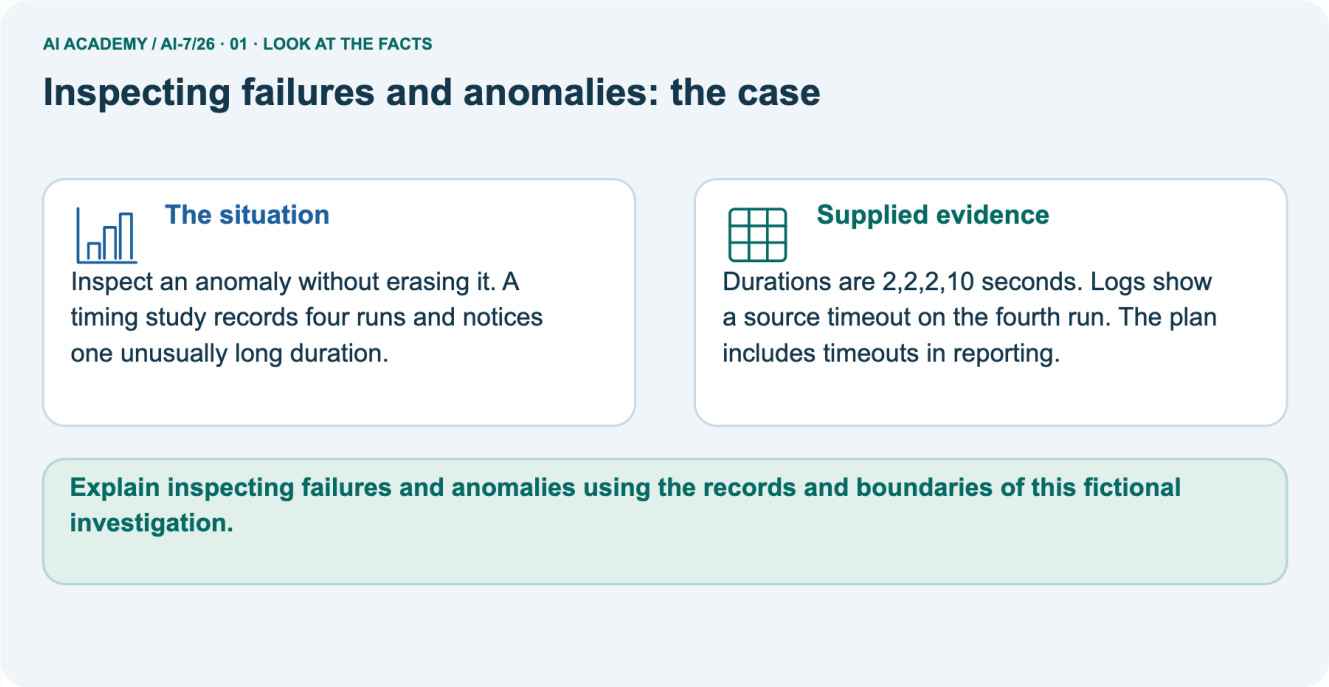
▼ **Reference · Six-picture concept journey**

REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Inspecting failures and anomalies: the case



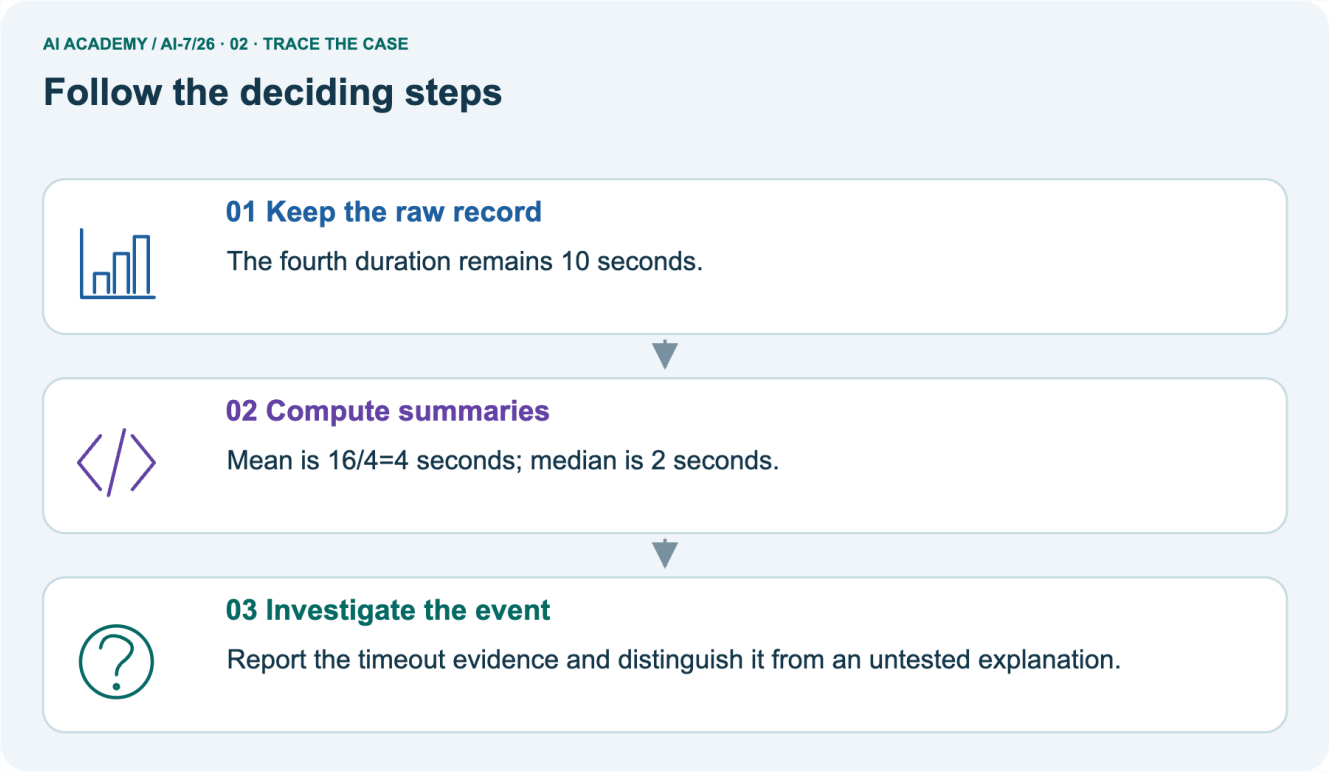
Follow the labelled picture with the matching explanation. Open the vector diagram

Inspect an **anomaly** without erasing it. A timing study records four runs and notices one unusually long duration. Supplied facts: Durations are 2,2,2,10 seconds. Logs show a source timeout on the fourth run.

The plan includes timeouts in reporting.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

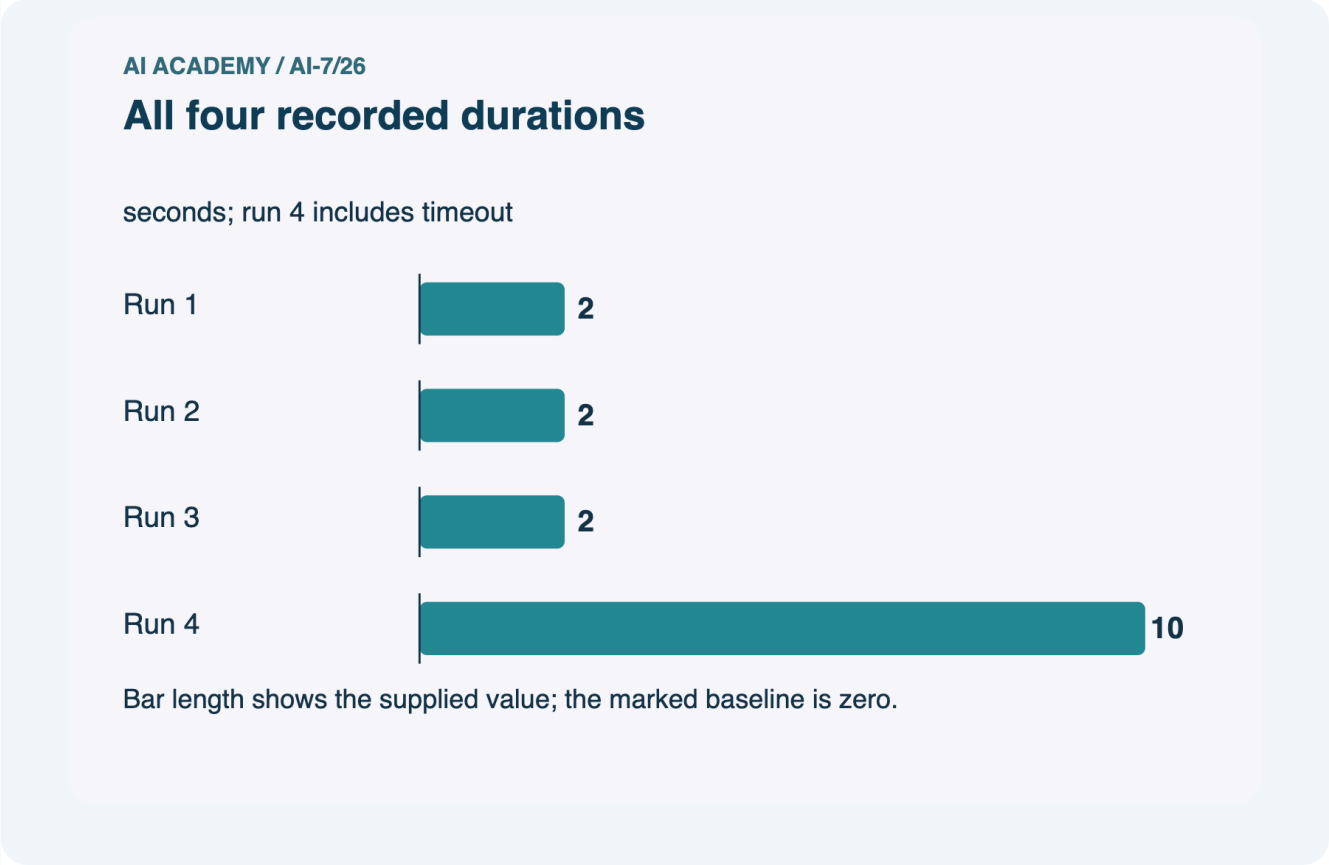


Follow the labelled picture with the matching explanation. Open the vector diagram

Keep the raw record: The fourth duration remains 10 seconds. → Compute summaries: Mean is 16/4=4 seconds; median is 2 seconds. → Investigate the event: Report the timeout **evidence** and distinguish it from an untested explanation.

Point to each numbered stage and explain which recorded **input** it uses.

03. All four recorded durations

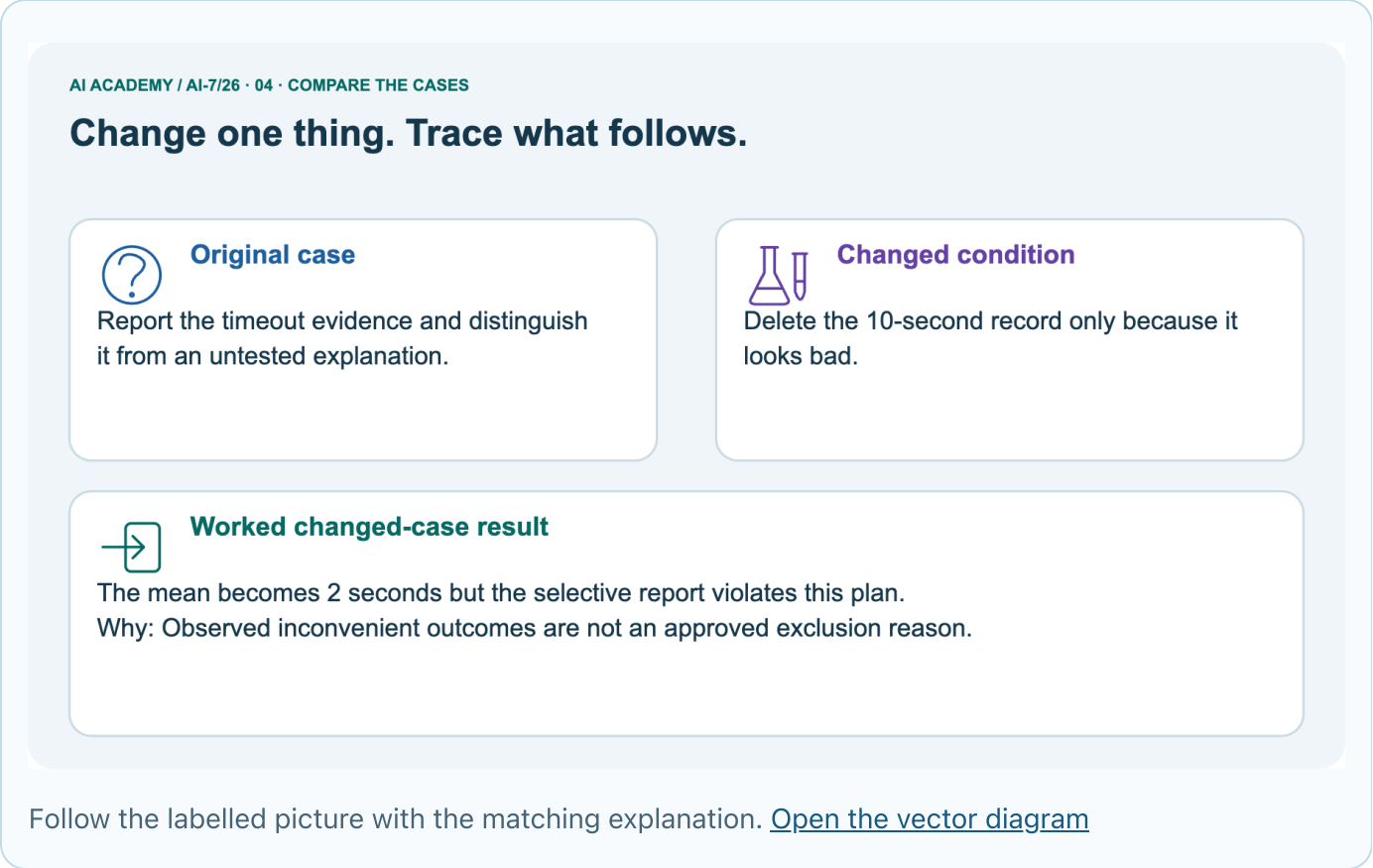


Follow the labelled picture with the matching explanation. Open the vector diagram

All four recorded durations. Units: seconds; run 4 includes timeout. Run 1: 2. Run 2: 2. Run 3: 2. Run 4: 10.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



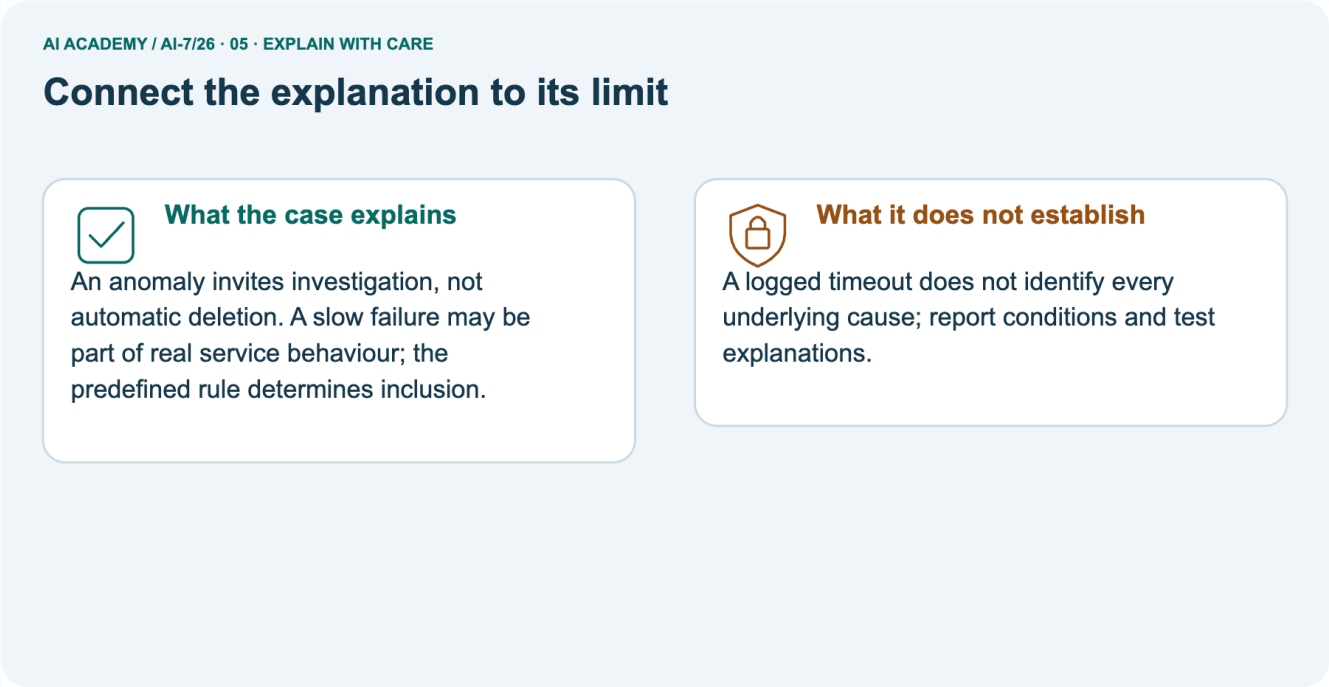
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Report the timeout **evidence** and distinguish it from an untested explanation.
Changed condition: Delete the 10-second record only because it looks bad.

Expected result: The mean becomes 2 seconds but the selective report violates this plan.
Reason: Observed inconvenient outcomes are not an approved exclusion reason.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



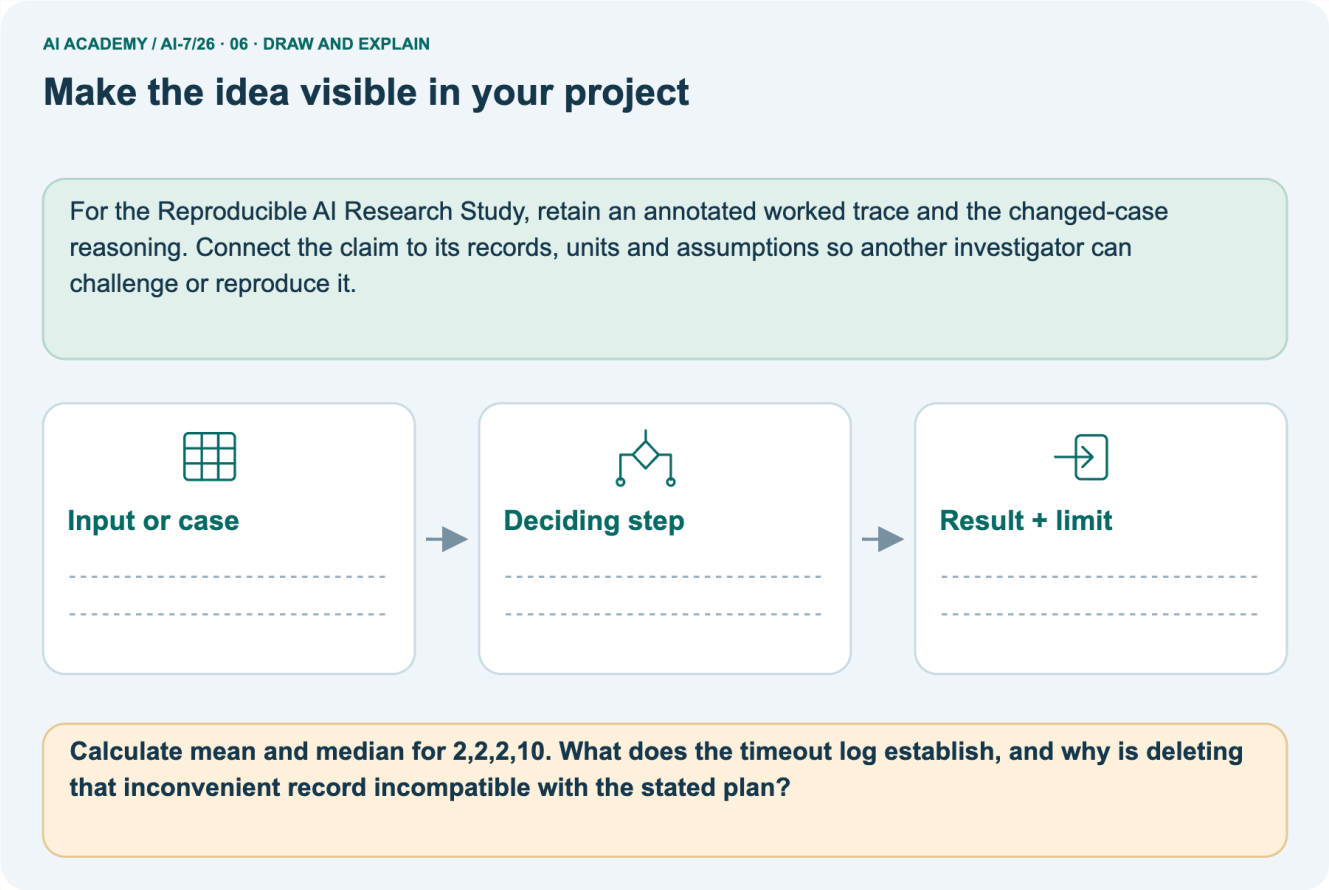
Follow the labelled picture with the matching explanation. Open the vector diagram

An **anomaly** invites investigation, not automatic deletion. A slow failure may be part of real service behaviour; the predefined rule determines inclusion.

Limit: A logged timeout does not identify every underlying cause; report conditions and test explanations.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Calculate mean and median for 2,2,2,10. What does the timeout log establish, and why is deleting that inconvenient record incompatible with the stated plan?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Inspecting failures and anomalies: the case

02. Follow the deciding steps

03. All four recorded durations

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.